STAT 527 Advanced Regression Analysis
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Estimation in Multiple Linear Regression

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Multiple Regression

So far we have only considered simple linear regression (only one predictor). More generally, consider several predictors ...

Let X_1, X_2, ..., X_p be p predictors of a response Y.

The data values might be listed as

$$\begin{array}{cccccc} y_1 & x_{11} & x_{12} & \cdots & x_{1p} \\ y_2 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_n & x_{n1} & x_{n2} & \cdots & x_{np} \end{array}$$

Model, Parameters, Assumptions

The **model equation** is

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \qquad i = 1, \ldots, n$$

The parameters $\beta_0, \beta_1, \beta_2, \ldots, \beta_p$ are the **coefficients**:

β_0 is the intercept and $\beta_1, \beta_2, \ldots, \beta_p$ are *partial* slopes

The same **assumptions** on errors $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ as in simple linear regression:

- zero mean $(\mathrm{E}(\varepsilon_i) = 0)$
- uncorrelated $(\mathrm{cov}(\varepsilon_i, \varepsilon_j) = 0,\ i \neq j)$, and
- homoscedastic $(\mathrm{var}(\varepsilon_i) = \sigma^2$ does not depend on $i)$.

The **error variance** is σ^2.

Notation

Let

$$\boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \qquad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix}$$

$$\boldsymbol{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

Then the regression equations for the n cases may be written together as

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Fitted values and Residuals

Let p' denote the number of columns of $\boldsymbol{X}$. For problems with an intercept, $p' = p + 1$. (With no intercept, $p' = p$.)

Given a sample of data, we wish to obtain an estimator of $\boldsymbol{\beta}$:

$$\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_p)^T$$

With respect to this estimator, we can define the vector of **fitted values** (or **predicted values**)

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}}$$

and the vector of **residuals**

$$\hat{\boldsymbol{\varepsilon}} = \boldsymbol{y} - \hat{\boldsymbol{y}}$$

Least Square Residuals

Recall that the residuals $\hat{\varepsilon}_i$ are defined by

$$\hat{\varepsilon}_i = y_i - \hat{y}_i, \qquad\qquad i = 1, 2, \ldots, n$$

and they form a vector $\hat{\boldsymbol{\varepsilon}} = \boldsymbol{y} - \hat{\boldsymbol{y}}$.

When the model is fit using least squares, these are also called the **least-squares residuals**.

When the model has an intercept, the least squares residuals satisfy

$$\sum_{i=1}^{n} \hat{\varepsilon}_i = 0$$

Why? See next slide

Answer

From the normal equation, we see that

$$\hat{\epsilon}^\top X = 0.$$

In particular, when there is an intercept, the first column of X are 1's, and it follows that

$$\sum_{i=1}^{n} \hat{\epsilon}_i = 0.$$

Least Squares

To generalize least-squares estimation from simple linear regression to multiple regression:

We want an estimator $\hat{\boldsymbol{\beta}}$ that minimizes the sum of squared residuals

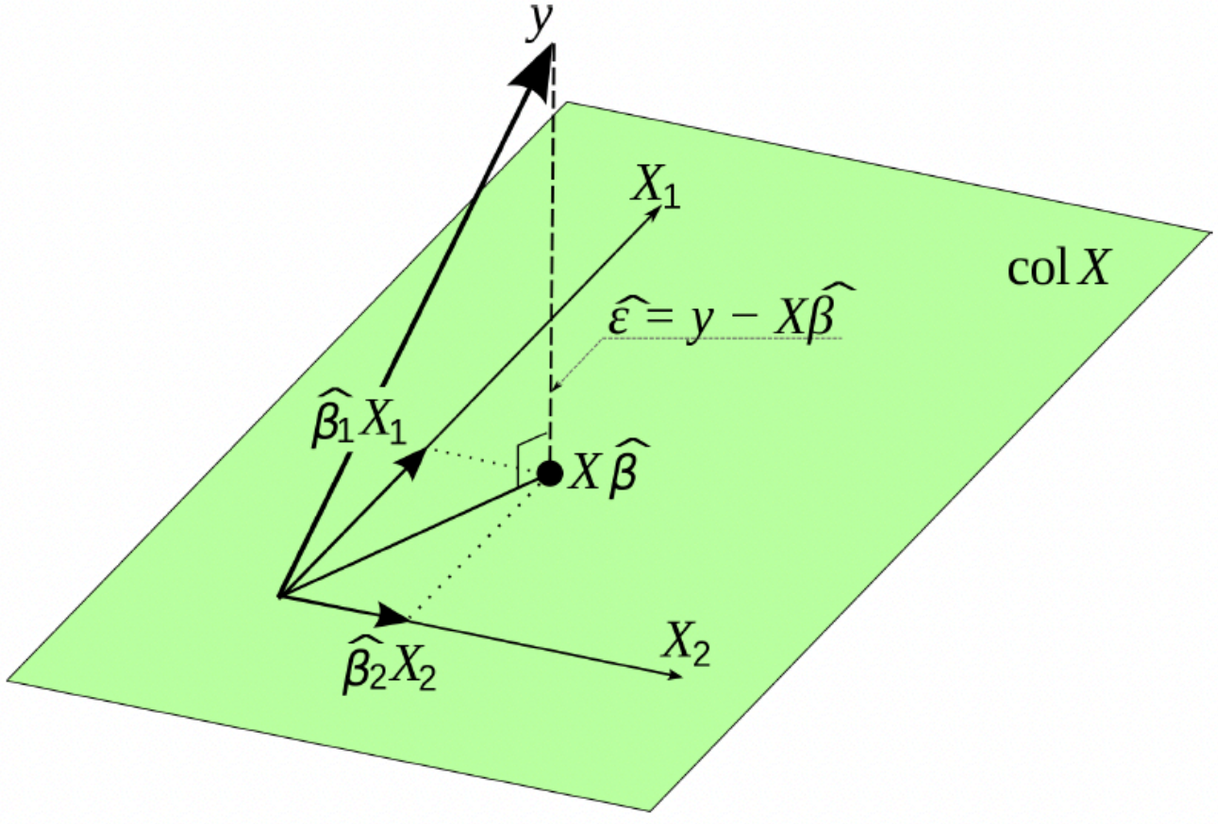
$$\sum_{i=1}^{n} \hat{\varepsilon}_i^2 \quad = \quad \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Notice: This is the same thing as minimizing the *Euclidean distance* between the vectors $\boldsymbol{y}$ and $\hat{\boldsymbol{y}}$.

Projection

Note that $\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}}$ can only range over vectors in the subspace spanned by the columns of $\boldsymbol{X}$.

Thus, to minimize the distance between $\boldsymbol{y}$ and $\hat{\boldsymbol{y}}$, it must be that $\hat{\boldsymbol{y}}$ is the *projection* of $\boldsymbol{y}$ onto the space spanned by the columns of $\boldsymbol{X}$.



Least Squares Estimator

Intuitively, it is obvious that this requires $\hat{\boldsymbol{\varepsilon}} = \boldsymbol{y} - \hat{\boldsymbol{y}}$ to be *orthogonal* to every column of $\boldsymbol{X}$.

That is, this *projection* is the (unique) vector $\hat{\boldsymbol{y}}$ satisfying

$$\boldsymbol{X}^T(\boldsymbol{y} - \hat{\boldsymbol{y}}) = \boldsymbol{0}$$

Thus,

$$\boldsymbol{X}^T\boldsymbol{y} = \boldsymbol{X}^T\boldsymbol{X}\hat{\boldsymbol{\beta}}$$

and

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$$

The vector $\hat{\boldsymbol{\beta}}$ is the **least squares estimator** of $\boldsymbol{\beta}$.

Review: Projection Matrices

- A square matrix P is a **projection** if it is symmetric and $P^2 = P$ (what if you drop the symmetric assumption? Consider rank-1 case).
- Equivalently, $P^\top (I - P) = 0$.
- By the spectral decomposition of symmetric matrices, there exist orthogonal Q such that $P = Q^\top \Lambda Q$, where

$$\Lambda = \begin{bmatrix} I & O \\ O & O \end{bmatrix}$$

  is a diagonal matrix.
- $X(X^\top X)^{-1} X^\top$ is a projection.

Thank you!

We gratefully acknowledge the former course instructor Naveen Narisetty and online resources for some of the images and slides presented.